

Dart Practice Task: Simple Library Management System

Goal:

Create a Dart program that practices basic programming and OOP concepts.

Requirements:

1. Create a Book class:

- Properties:

- title

- author

- isAvailable

- Create a constructor.

- Add a method displayInfo() to print book details.

2. Create an abstract class Person:

- Property:

- name

- Method:

- showRole()

3. Create a Member class that extends Person:

- Property:

- memberId

- Create a list of borrowed books.

- Override showRole() and print:

"I am a library member"

4. Create a Library class:

- Contains a list of books.
- Add these methods:
- addBook(Book book)
- showBooks()
- borrowBook(String title, Member member)
- returnBook(String title, Member member)

5. In main():

- Create several books.
- Create a member.
- Add books to the library.
- Borrow a book.
- Return a book.
- Display the final library status.

Concepts to Practice:

- Variables and data types
- Conditions
- Loops
- Lists
- Functions
- Classes and objects
- Constructors
- Encapsulation
- Inheritance
- Polymorphism
- Abstract classes

- Exception handling